Result 1 (underlying point belongs). [Mathlib, Subtype.property]
Let X be a type. Let S be a subset of X. Let x be an element of S. Then

$$x \text{ belongs to } S.$$

Adapted from the cited source.

Result 2 (reversing a path twice). [Mathlib, Path.symm_symm]
Let X be a type. Let t be a topology on X. Let x and y be elements of X. Let p be a path from x to y in X with topology t. Then

$$(p^{-1})^{-1} = p.$$

Adapted from the cited source.

Result 3 (constant paths exist). [Mathlib, Path.refl]
Let X be a type. Let t be a topology on X. Then

$$(\text{for every } x : X,\ (\text{there exists } p : \mathrm{Path}_t(x, x),\ p = \mathrm{constant}(x)))\,.$$

Adapted from the cited source.

Result 4 (paths can be joined). [Mathlib, Path.trans]
Let X be a type. Let t be a topology on X. Let x, y and z be elements of X. Let p be a path from x to y in X with topology t. Let q be a path from y to z in X with topology t. Then

$$(\text{there exists } r : \mathrm{Path}_t(x, z),\ r = p * q)\,.$$

Adapted from the cited source.

LLM-assisted Clean specimens and proofs (gpt-6, 2026-11-06). Mathematical sources are attributed above; see PROVENANCE.md.